

Design and Implementation of an Explainable AI-Assisted Skin Lesion Pre-Screening System Using Pretrained Deep Learning Models for Low-Resource Health Care Settings in Rwanda

BSc. in Software Engineering

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List of Acronyms and Abbreviations

AI – Artificial Intelligence

API – Application Programming Interface

CDSS – Clinical Decision Support System

CNN – Convolutional Neural Network

ERD – Entity-Relationship Diagram

GCGO – Grand Challenges Global Opportunities

Grad-CAM – Gradient-weighted Class Activation Mapping

ISIC – International Skin Imaging Collaboration

MC Dropout – Monte Carlo Dropout

ML – Machine Learning

MoH – Ministry of Health

REST – Representational State Transfer

ROC-AUC – Receiver Operating Characteristic – Area Under Curve

SDLC – Software Development Life Cycle

UML – Unified Modelling Language

XAI – Explainable Artificial Intelligence

Abstract

Skin diseases are a leading cause of non-fatal illness worldwide, but in sub-Saharan Africa, access to dermatology care is limited. In Rwanda, there are few specialists. The first dermatology residency began in 2023. There are about 0.1 doctors per 1,000 people, and most rural areas lack dermatology services. Early visual screening at the primary care level could help with triage, but no easy-to-use or transparent decision-support tool exists for non-specialist healthcare workers. This project aims to design and build a web-based, AI-assisted skin lesion pre-screening system using pretrained deep learning models and public dermatology datasets. The system will allow clinicians to upload a skin lesion image and receive a preliminary classification, a confidence score, a visual explanation of the model's decision, and a referral recommendation while still relying on their clinical judgment. The system will be evaluated using technical metrics like precision, recall, F1-score, balanced accuracy, ROC-AUC, and calibration, as well as mission-oriented metrics such as uncertainty flagging rate, melanoma sensitivity, and usability score. The model is expected to achieve a balanced accuracy of about 88–93% on the HAM10000 benchmark, keep the melanoma false-negative rate below 15%, and reach a mean System Usability Scale score above 65. This project aims to meet Rwanda's need for better early skin health assessment by showing that an explainable, responsible AI screening tool can be built entirely from open-source resources.

CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

Skin diseases are a significant and often overlooked health problem worldwide. According to the Global Burden of Disease Study, skin conditions are among the top causes of non-fatal disability, affecting more than one billion people at any time. Despite this, dermatology is one of the least supported medical specialties in low- and middle-income countries, where health systems already face many challenges. In sub-Saharan Africa, there are many skin conditionsincluding infectious diseases, neglected tropical diseases with skin symptoms, and cancersbut very few specialist dermatologists. As a result, most skin diseases are either not detected or are treated by clinicians without dermatology training.

Rwanda exemplifies this problem. The country has made major health improvements over the last twenty years, but the number of dermatology specialists remains very low. The first formal dermatology training program started only in 2023. There are about 0.1 physicians per 1,000 people, far below the WHO's recommended minimum of 1 per 1,000. Most specialists are based in Kigali, so people in rural areas have little access to skin health assessment. This gap is critical because many skin diseases, including serious ones, can be recognized early by visual inspection. If this early check does not happeneither due to lack of clinicians or insufficient experiencethe chance for early treatment is missed.

To deal with the shortage of specialists in low-resource health systems, two main approaches have been used: increasing the workforce and task shifting. Expanding the workforce through training and international recruitment is a long-term solution and cannot meet immediate screening needs. Task shifting, which allows less specialized health workers to do some tasks usually done by specialists, has worked in areas like HIV care and maternal health. However, it relies on having good tools to help these workers make decisions outside their main training. In dermatology, such tools have not been widely used in sub-Saharan Africa.

Recent advances in artificial intelligence, especially deep learning for medical images, have created a new option: AI-assisted decision support. Over the last ten years, studies have shown that deep learning models can classify skin lesion images into categories with accuracy similar to trained dermatologists, using public datasets like HAM10000 and the ISIC archive. These results are promising, but turning this research into practical tools for low-resource settings has not yet happened. Current commercial and research systems have several problems: most are proprietary

and require expensive licenses; they are trained mostly on images of lighter skin, raising fairness concerns for African populations; they do not provide visual explanations to help non-specialists understand results; and they do not show uncertainty, so they may give confident answers even when the image is unclear.

These are not minor technical issues; they are major barriers to using AI tools responsibly in Rwanda. If a screening tool cannot explain its decisions, clinicians cannot trust it. If it does not show when it is unsure, it cannot be safely used for triage. If it is trained mostly on lighter skin, it may not work well for Rwandan patients. The main problem is not just the lack of AI dermatology tools but the lack of tools designed from the start to be explainable, show uncertainty, work fairly across skin tones, and be usable in low-resource settings. This study aims to fill that gap by designing, building, and testing a web-based, explainable, uncertainty-aware AI skin lesion pre-screening system for Rwanda. Most specialists are based in Kigali, so most Rwandans with skin conditions are seen by general practitioners or community health workers without dermatology training. There is no systematic screening process to help these clinicians decide which skin problems need further investigation. As a result, many skin diseasesincluding those detectable early by visual inspectionare missed at the primary care level (Ministry of Health Rwanda, 2023; World Bank, 2019; Doximity Foundation, 2023).

In clinical practice, the challenge is not just access but also decision support. Even when a trained clinician is available, telling which lesions need specialist follow-up requires pattern-recognition skills that take years to develop. Without a support tool, clinicians have to rely only on their own visual judgment for important decisions. There is also no way to record, track, or review how lesions change over time, so each visit stands alone. A tool that offers structured preliminary classification, a visual explanation of its reasoning, and a clear confidence score could greatly improve the information clinicians have when making triage decisions.

Current AI-assisted dermatology software does not solve this problem. Commercial tools are proprietary and out of reach for public healthcare facilities with limited budgets. Both commercial and research systems are mostly trained on images of lighter skin types (Fitzpatrick I–III), and studies show they are much less accurate for people with darker skina fairness issue for Rwanda (Daneshjou et al., 2022; Groh et al., 2024; Kinyanjui et al., 2020). Importantly, no available system offers both explainability (showing how the model made its decision) and

uncertainty measurement (indicating when the model is unsure and human review is needed) in an open-source, web-based format for non-specialist users in low-connectivity settings. This project addresses this gap by designing and building a system that includes all these features: preliminary lesion classification, Grad-CAM visual explanation, Monte Carlo Dropout uncertainty assessment, structured referral guidance, and a web-based interface that does not require specialist infrastructure.

1.3 Project's Main Objective

The development of a web-based skin lesion pre-screening tool should involve accepting an uploaded image and then providing a preliminary classification of the lesion, together with a visual explanation of the decision behind the classification, a confidence score, and a referral recommendation this will allow non-specialist healthcare workers in resource-limited settings to make more informed triage decisions.

1.3.1 Specific Objectives

1. In order to determine the system requirements, it is necessary to look at current solutions, and to obtain and get ready a suitable dataset of skin lesions that is publicly available for use in an initial multi-class classification.
2. Constructing a web-based pre-screening system which offers initial lesion classification, a visual explanation of the predictions, referral advice flagged for uncertainty, and a case history log.
3. We need to assess whether the system provides reliable classification performance, meaningful explainability, and an output presentation that is usable for non-specialist users.

1.4 Research Questions

The following research questions derive from the three objectives:

- For the first research question, how accurately can a deep learning model that has been pretrained classify the different categories of skin lesions using the HAM10000 public

dataset, and what is the difference in performance between a lightweight baseline model and the more advanced model that has been proposed?

- RQ2: Is it possible to incorporate Gradient-weighted Class Activation Mapping and Monte Carlo Dropout's uncertainty assessment into a web-based pre-screening system in order to provide non-specialist users with transparent and uncertainty-aware classification support?
- RQ3: Does the system in question manage to achieve acceptable levels of classification performance, provide meaningful explainability, and offer good usability in the context of a prototype intended for use in low-resource healthcare settings?

1.5 Project Scope

The project involves the creation of an academic prototype rather than a product that is ready for clinical use and will not be employed in real patient care.

- Classification covers only the seven lesion types in HAM10000. These are melanocytic nevi, melanoma, benign keratosis-like lesions, basal cell carcinoma, actinic keratoses, vascular lesions, and dermatofibroma.
- The training data is taken from the publicly available HAM10000/ISIC archive and no patient images will be collected in Rwanda.
- The usability evaluation will use a small group based in Kigali (consisting of peers, academic supervisors, or healthcare students) not practicing clinicians.
- The system won't recommend a treatment, confirm a diagnosis, or grant clinical approval; all of its outputs come with an explicit disclaimer.
- The application of this in real life would need clinical validation, ethical approval from the appropriate health authorities in Rwanda, compliance with data privacy regulations, and regulatory scrutiny all of these having been designated as part of future work.

1.6 Significance and Justification

If successful, this prototype will show that a technically strong, explainable, and uncertainty-aware AI pre-screening tool can be built entirely from open-source resources and

public datasets by a single undergraduate student. This provides a model that others can follow for future projects in Rwanda and similar low-resource settings.

This project directly addresses Rwanda's need for better early skin health assessment and follows the Ministry of Health's National Digital Health Strategic Plan (2018–2023), which highlights clinical decision-support tools and telemedicine as priorities. By making explainability, uncertainty communication, and responsible AIincluding skin tone bias and domain shiftcore design requirements, the project shows an approach to healthcare AI that values equity and safety as much as technical performance.

1.7 Research Budget

Python, PyTorch, torchvision	0.00	Open-source
FastAPI, React.js, PostgreSQL	0.00	Open-source
HAM10000 / ISIC dataset (CC BY-NC 4.0)	0.00	Publicly available
pytorch-grad-cam, SQLAlchemy	0.00	Open-source
GitHub (free tier)	0.00	Version control
Google Colab Pro (optional GPU)	~50.00	Only if local GPU insufficient
Cloud hosting – Render free tier	0.00	Prototype deployment

Total	~50.00 USD	

Table 1.1: Research Budget

All the tools, frameworks, and datasets are open source and can be obtained publicly at no cost; the only possible cost, amounting to about USD 50, is optional cloud GPU time should it turn out that the local hardware is not sufficient for training the model.

1.8 Research Timeline

1	Literature Review & Problem Definition	1–3
2	Dataset Acquisition & Investigation	1-3
3	Requirements Analysis & System Design	4
4	Data Pre-processing Pipeline	5-6
5	Baseline Model Implementation	6-7
6	Proposed Model Training & Comparison	6-8

7	XAI & Uncertainty Layer Integration	7-8
8	Backend API Development	7-8
9	Frontend Development	9
10	System Integration & Testing	9
11	Evaluation & Metrics Collection	9
12	Documentation & Final Presentation	9

Table 1.2: Research Timeline

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

The review looked at the software literature covering AI-assisted skin lesion classification, explainable AI in medical imaging, uncertainty assessment in deep learning, clinical decision support systems, and AI fairness in dermatology. Searches were carried out on PubMed, IEEE Xplore, ACM Digital Library, arXiv, and Google Scholar using the following terms: 'skin lesion classification deep learning', 'Grad-CAM medical imaging', 'Monte Carlo dropout uncertainty', 'clinical decision support dermatology', 'AI fairness skin tone HAM10000', and 'digital health low-resource Rwanda'. Publications from the period 2015 to 2026 were given priority, and where

appropriate earlier foundational works were also included. About 55 sources were identified; the most relevant of these are summarized below.

2.2 Overview of Existing Systems

Google DermAssist (Google Health) is a deep-learning skin condition assessment tool which has been commercially developed. A peer-reviewed assessment, based on a varied collection of clinical images, showed significant differences in performance according to Fitzpatrick skin types, the accuracy being much lower for types V to VI than for types I to III. The model is proprietary and closed-source, so it can only be accessed with a licence, and it gives suggestions as to the condition without providing any visual explanation of the reasons for its decision (Daneshjou et al., 2022).

SkinVision is a commercial mobile application which provides a risk assessment for lesions that have been photographed, aiming at enabling consumers to carry out self-assessment in the European markets. The application is mainly trained on populations with lighter skin, does not provide any visual attribution, and because of both its geographic and commercial approach it is not suitable for use in Rwanda's public healthcare system.

SkiNet, as described by Mahbod et al. in 2022, is a research prototype which integrates lesion segmentation, classification, and uncertainty assessment and is tested using the HAM10000 dataset. It is technically strong but is available only as a research codebase and does not include a deployed clinical interface, an API layer, or a database; furthermore, it has not been assessed for usability by non-specialists or for applicability among sub-Saharan African populations.

MelanomaNet (2025) is a research system that has recently been developed; it uses an EfficientNet backbone together with Grad-CAM++ visualization and Monte Carlo Dropout for the purpose of measuring uncertainty on the ISIC 2019 dataset. The system attains good results in terms of technical performance and represents the closest published example to the one proposed here. Yet it does not include a complete full-stack deployment (it has no API, no database, and no production frontend) and provides no analysis regarding fairness or domain-shift with respect to skin-tone diversity.

2.3 Strengths and Weaknesses of Existing Systems

Together, the systems under review show that it is technically possible to carry out skin lesion classification using deep learning, that Grad-CAM methods for explaining results and MC Dropout for assessing uncertainty can be used with CNN-based classifiers for medical imaging, and that the EfficientNet family of architectures is currently the strongest available pre-trained model family for this task. The systems that have been identified, however, all have one or more of the following weaknesses, and these are the ones that the proposed system is intended to overcome:

- In resource-constrained environments within the public sector, proprietary or closed architectures cannot be accessed.
- There is no visual explainability since the outputs are simply labels or risk categories without indicating which image features led to the decision.
- There is no assessment of uncertainty: high-confidence outputs are generated even when the input is ambiguous or atypical, without any procedure to indicate such cases for compulsory human review.
- Training data sets include fewer examples of darker Fitzpatrick skin types (V and VI). Studies show that these sets do not work as well for people from sub-Saharan Africa (Daneshjou et al., 2022. Groh et al., 2024. Kinyanjui et al., 2020).
- Incomplete software systems: research prototypes do not have deployed API layers, databases, and clinician-facing interfaces.
- There was no regard for the limitations imposed by scarce connectivity or for the medical procedures carried out by people who are not specialists.

2.4 A review of relevant work

The HAM10000 dataset, which was introduced in 2018, includes 10,015 dermoscopic images distributed among seven diagnostic categories and has thus become the standard against which multi-class skin lesion classification is measured. It is known that the dataset has a severe class imbalancewith melanocytic nevi making up 66.9% and dermatofibroma only 1.1% as well as the

fact that it was collected from two areas that are mainly light-skinned; these are recognised as limitations of the project's methodology.

A study involving fundamental research, which was published in Nature in 2017, showed that a convolutional neural network trained on more than 100,000 clinical skin images was able to classify skin cancer with accuracy at the level of that of a dermatologist, thus proving that deep learning can form a credible foundation for AI-assisted dermatology and prompting the creation of openly available reference datasets.

Gradient-weighted Class Activation Mapping (Grad-CAM), which was proposed in 2017, allows for post hoc visual explanations to be obtained for any CNN architecture without the need for any changes to the architecture. It generates spatial heatmaps which show the image regions that had the greatest influence on the classification decision, and has therefore become the standard method of explanation in deep learning applications in medical imaging because of its efficiency and the way it is understandable to clinical audiences (Selvaraju et al., 2017).

Monte Carlo Dropout, which was formalised in 2016 as a practical Bayesian approach to dealing with uncertainty in deep neural networks, estimates both epistemic and aleatoric uncertainty by carrying out a number of stochastic forward passes with dropout remaining active when performing inference. In the case of clinical imaging, the entropy score obtained in this way offers a sound way of identifying low-confidence predictions for human examination rather than treating them as final outputs (Gal & Ghahramani, 2016; Leibig et al., 2017).

Several fairness studies have shown that classifiers trained on the HAM10000 and ISIC datasetsdatasets that primarily include Fitzpatrick types I to IIIachieve considerably less accuracy when dealing with types V and VI. These findings form the empirical foundation for the domain shift issue that is at the heart of this project's responsible AI analysis (Kinyanjui et al., 2020; Daneshjou et al., 2022; Groh et al., 2024).

A study that has combined EfficientNet-family backbone networks with Grad-CAM++ for explainability and MC Dropout for uncertainty, using the HAM10000 dataset, has reached an accuracy of up to 96% and a ROC-AUC of 99% when uncertainty-filtered inference is applied, showing that the three components can be incorporated into a single classification pipeline and achieve high technical performance (Olubunmi et al., 2025). This research serves as the main methodological reference for the XAI and uncertainty aspects of the proposed system.

2.5 A general comment and conclusion

The relevant literature shows that each of the individual technical elements of the proposed systemnamely transfer learning, Grad-CAM explainability, MC Dropout for uncertainty assessment, and RESTful full-stack integration has already been thoroughly validated by previous research. The issue here is not one of algorithmic approach but rather one of integration and of context. There is no current system that brings together all four of these components to produce a complete, deployed, open-source tool that is designed for use by people who are not specialists in the field and that is tailored to a low-resource sub-Saharan African healthcare setting. The problem of skin tone representation also introduces an ethical aspect which no currently deployed tool has adequately dealt with for the groups that this project is intended to serve. These two gaps integration and contextual responsibility make up the main research contribution of this project.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

The chapter sets out the research design, machine learning pipeline, system requirements, and system design for the AI-assisted skin lesion pre-screening prototype. An iterative prototype software development life cycle (SDLC) is used since the choice of model depends on the results of empirical baseline tests, the requirements are to some extent emergent, and the development process which involves designing, implementing, evaluating, and refining is repeated within each phase. Chapter 3 includes a discussion of the ML pipeline, the dataset, the model comparison, the assessment criteria, the functional and non-functional requirements, and all the system and UML design specifications.

3.2 Research Design

The reason why the iterative SDLC is chosen rather than the linear model is that the model comparison stagewhich is a key part of the researchcannot be fully planned ahead of time; the baseline results have an influence on the model selection and the strategy for fine-tuning.

The sequence of development follows standard practice in ML systems engineering: investigation of the literature and the dataset → definition of the requirements and design → data pre-processing → baseline model → proposed model → integration of XAI and uncertainty → backend API → frontend → integration → evaluation → documentation.

3.2.1 The dataset and a description of the dataset

The main dataset is HAM10000 (Human Against Machine using 10,000 training images), which is available under the CC BY-NC 4.0 licence from the ISIC Archive and includes 10,015 dermoscopic images distributed among seven classes:

Melanocytic Nevi	NV	6,705	66.9%
Melanoma	MEL	1,113	11.1%
Benign Keratosis-like Lesions	BKL	1,099	11.0%
Basal Cell Carcinoma	BCC	514	5.1%
Actinic Keratoses / Bowen's	AKIEC	327	3.3%

Vascular Lesions	VASC	142	1.4%
Dermatofibroma	DF	115	1.1%

Table 3.1: Distribution of the HAM10000 Class (Tschandl et al., 2018)

The main challenges are: (a) a severe imbalance between classes, which was dealt with by using class-weighted loss functions and data augmentation; (b) the fact that the dataset was mainly gathered from light-skinned Europeans, a limitation due to domain shift that is explicitly discussed in the responsible AI section of the evaluation; and (c) dermoscopic artifacts such as hair and ruler markings, which were reduced by means of pre-processing. The data will be divided in the ratio of 70% / 15% / 15% (for training / validation / test) using stratified sampling.

3.2.2 Baseline Model Description

MobileNetV2 (having been pre-trained on ImageNet) is used as the computational baseline; with approximately 3.4 million parameters and the use of depthwise separable convolutions it achieves competitive results in image classification while keeping the computational cost low. The backbone is employed but its classification head is substituted by a seven-class output layer; the weights of the backbone are kept fixed at first and the new head is trained in order to set a lower-bound benchmark for use in model comparisons.

3.2.3 Proposed Model Description

The model that has been proposed is EfficientNet-B3 (which has been pre-trained on ImageNet and contains about 12 million parameters), this choice being based on its compound scaling method which simultaneously optimises depth, width, and resolution (300×300 pixels). Several published evaluations indicate that the various EfficientNet models attain a balanced accuracy of between 91% and 96% when tested on HAM10000 using transfer learning. The entire network will be fine-tuned end-to-end with the use of a low learning rate. To apply Grad-CAM to the final convolutional block of EfficientNet-B3 the pytorch-grad-cam library will be used; MC Dropout (involving T=10 stochastic inference passes) will generate uncertainty scores based on

entropy; and if the predictions exceed a certain entropy threshold a mandatory alert for a professional review will be triggered.

Pre-training	ImageNet	ImageNet
Parameters	~3.4 M	~12 M
Input Resolution	224×224 px	300×300 px
Computational Cost	Low	Moderate
Role	Lower-bound benchmark	Primary model

Table 3.2: Baseline vs. Proposed Model

3.2.4 Metrics for Evaluation

The technical metrics include per-class and macro Precision, Recall, and F1-score; Balanced Accuracy; ROC-AUC (using the one-versus-rest approach); the Confusion Matrix; the Expected Calibration Error; and the average inference time.

Metrics with a mission: the rate at which uncertainty is flagged (that is, the proportion of predictions that exceed the entropy threshold); specific attention to melanoma recall (since this is the highest-risk class and has the lowest acceptable tolerance for false negatives); aiming for a System Usability Scale (SUS) score of at least 65; and a qualitative assessment of Grad-CAM faithfulness.

3.3 System Analysis

3.3.1 Functional Requirements

FR-01	User can upload a skin lesion image via the web interface	Must Have
FR-02	System validates image format, resolution, and size before processing	Must Have
FR-03	System classifies the image across all 7 HAM10000 lesion classes	Must Have
FR-04	System estimates prediction uncertainty and displays a confidence score	Must Have
FR-05	System generates a Grad-CAM heatmap overlay for the top predicted class	Must Have

FR-06	High-uncertainty predictions trigger a mandatory professional review alert	Must Have
FR-07	Case metadata is stored in PostgreSQL for case history retrieval	Must Have
FR-08	All functions are accessible via a documented REST API	Must Have
FR-09	Basic user authentication restricts access to authorised users	Must Have

Table 3.3: Functional Requirements

3.3.2 Non-Functional Requirements

- The system provides the full screening result within 10 seconds of the image being submitted.
- Usability: the average SUS score of the specified test group is 65 or higher.
- Reliability is ensured through graceful handling of errors caused by invalid images, oversized uploads, and timeouts.
- Security involves the use of hashed passwords and the retention of only minimal image data.
- The codebase is modular and well documented, with the deployment instructions available on public GitHub.

- In terms of safety and ethics, all outputs come with an explicit disclaimer stating that the results are for use as decision support only and not as a clinical diagnosis.

3.3.3 Proposed Model Diagram Customized to Mission

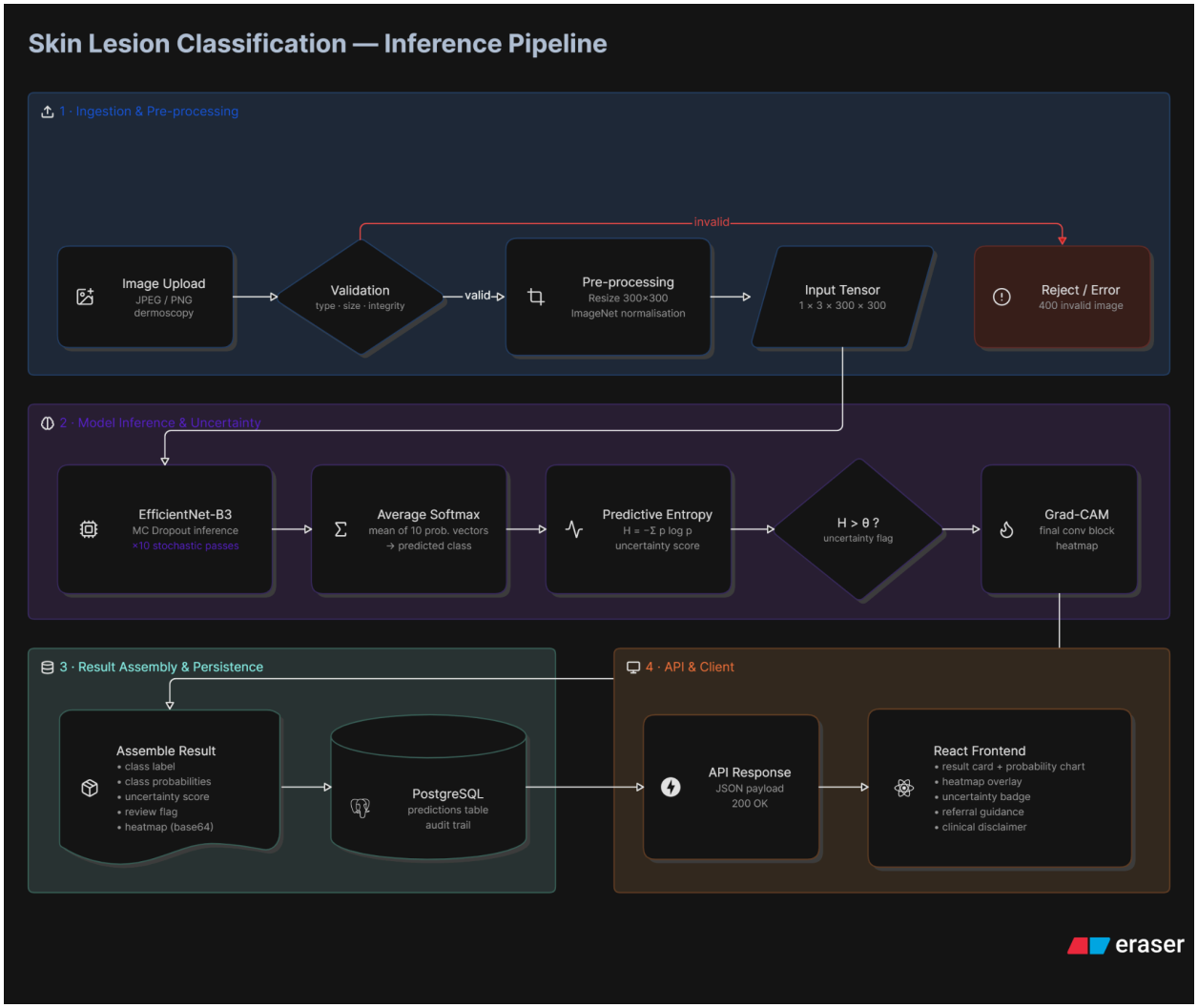
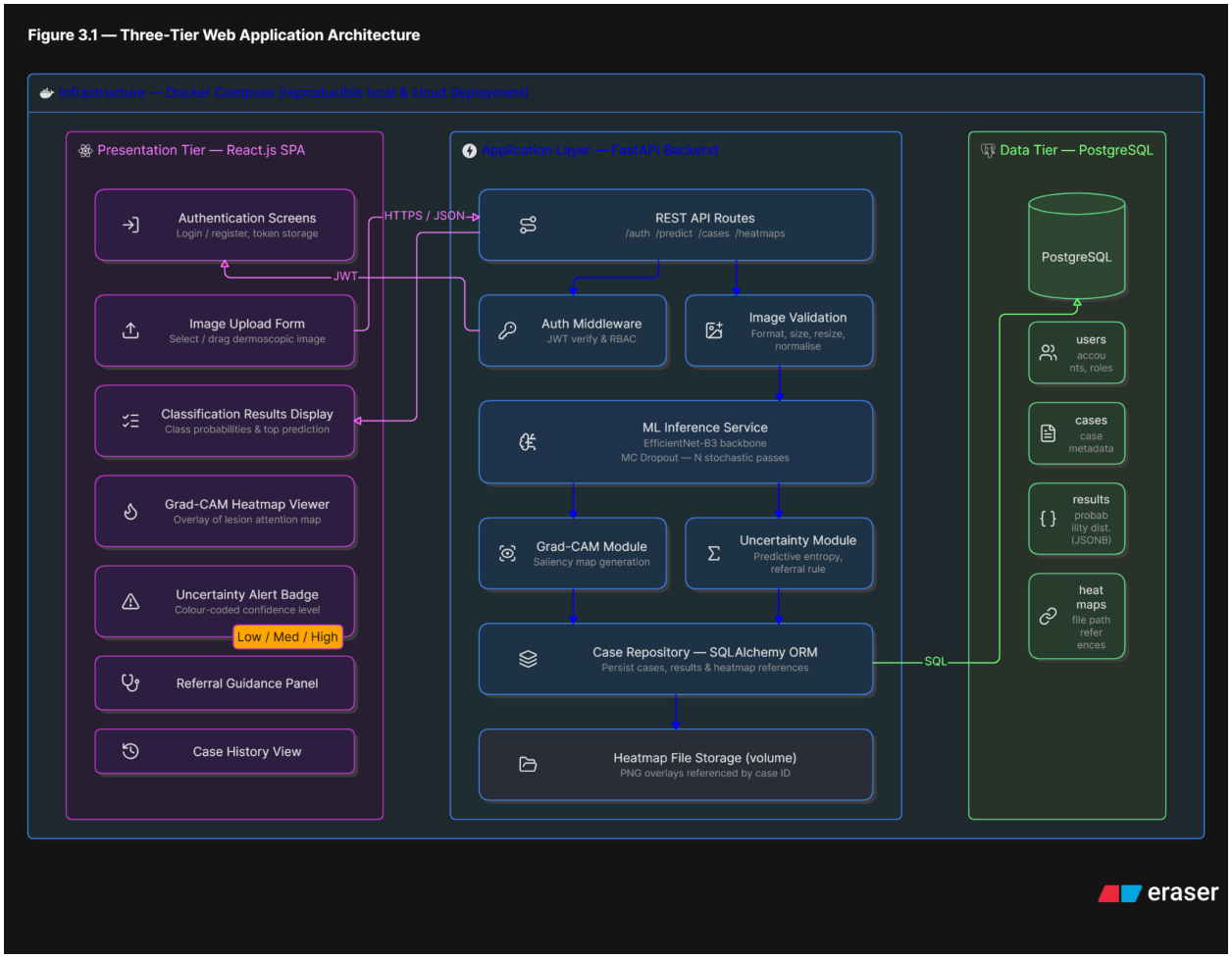


Figure 3.2 shows the end-to-end ML inference pipeline, having been adapted to correspond with the screening-support mission:

3.4 System Design

3.4.1 System Architecture



(Figure 3.1)

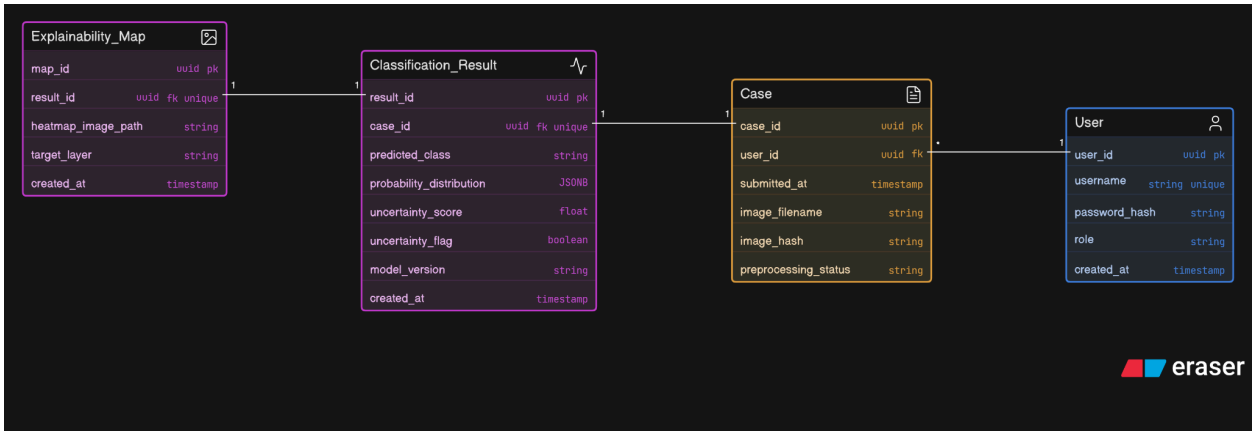
Three-tier web application:

- Presentation tier a React.js single-page application featuring an image upload form, the display of classification results, a Grad-CAM heatmap viewer, an uncertainty alert badge that uses colours to indicate the level of uncertainty, a panel providing referral guidance, and a case history view together with authentication screens.

- - Application layer: FastAPI backend, including authentication middleware (using JWT), image validation and pre-processing, the ML inference service (based on EfficientNet-B3 and MC Dropout), the Grad-CAM module, the uncertainty module, the case repository (using SQLAlchemy ORM), and the REST API routes.
- For the Data Tier PostgreSQL: user accounts, case metadata, the classification results (given as probability distributions in JSONB format), and references to heatmap files are stored.
- The infrastructure involves using Docker Compose to containerise all the services so that both local and cloud deployments can be reproduced.

3.4.2 Database Design / ERD (Figure 3.3)

Four entities with the following principal attributes and relationships:



3.4.3 UML Diagrams

All diagrams below are mandatory deliverables for the final dissertation and will be rendered with correct UML notation:

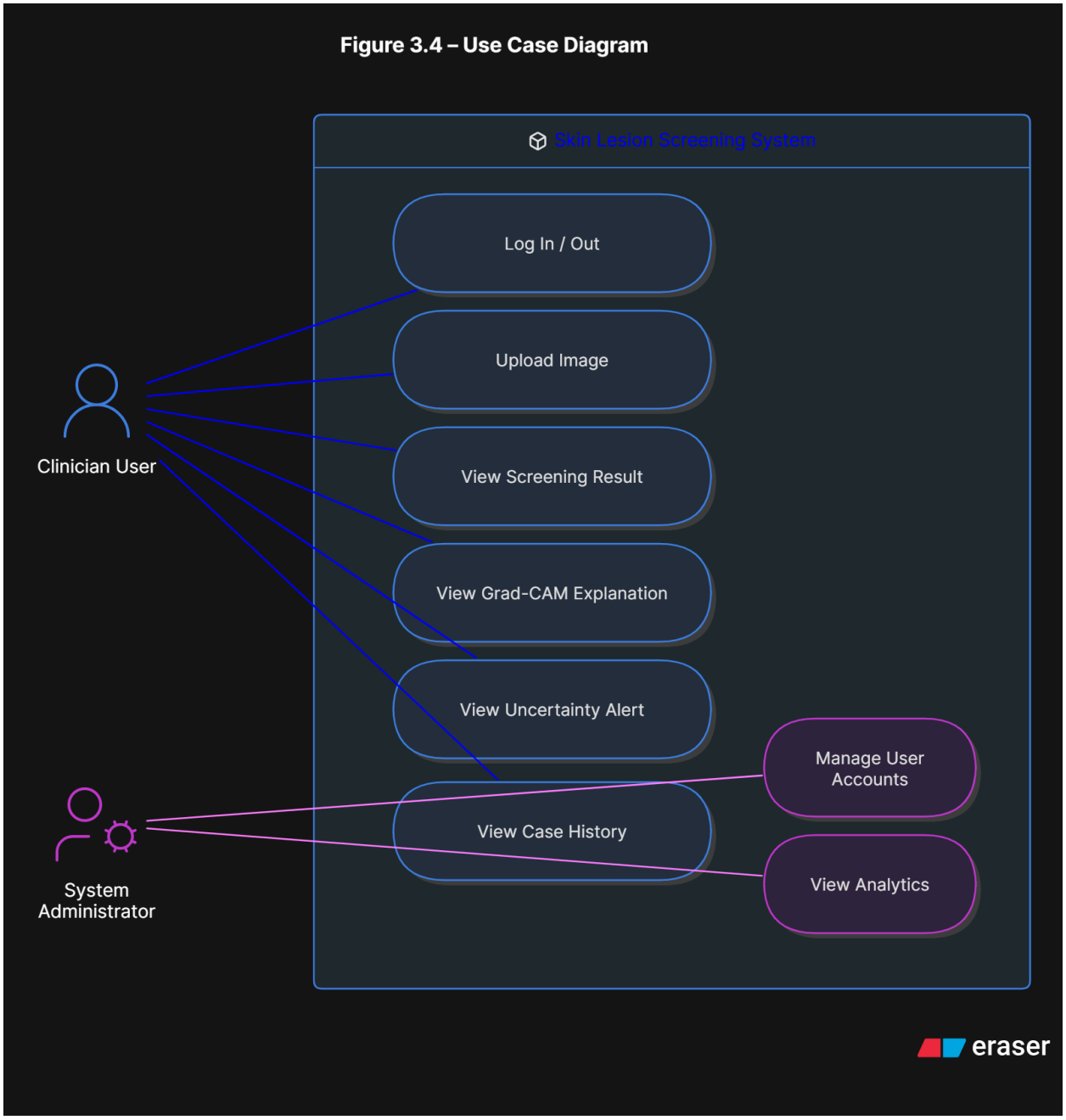


Figure 3.4 – Use Case Diagram

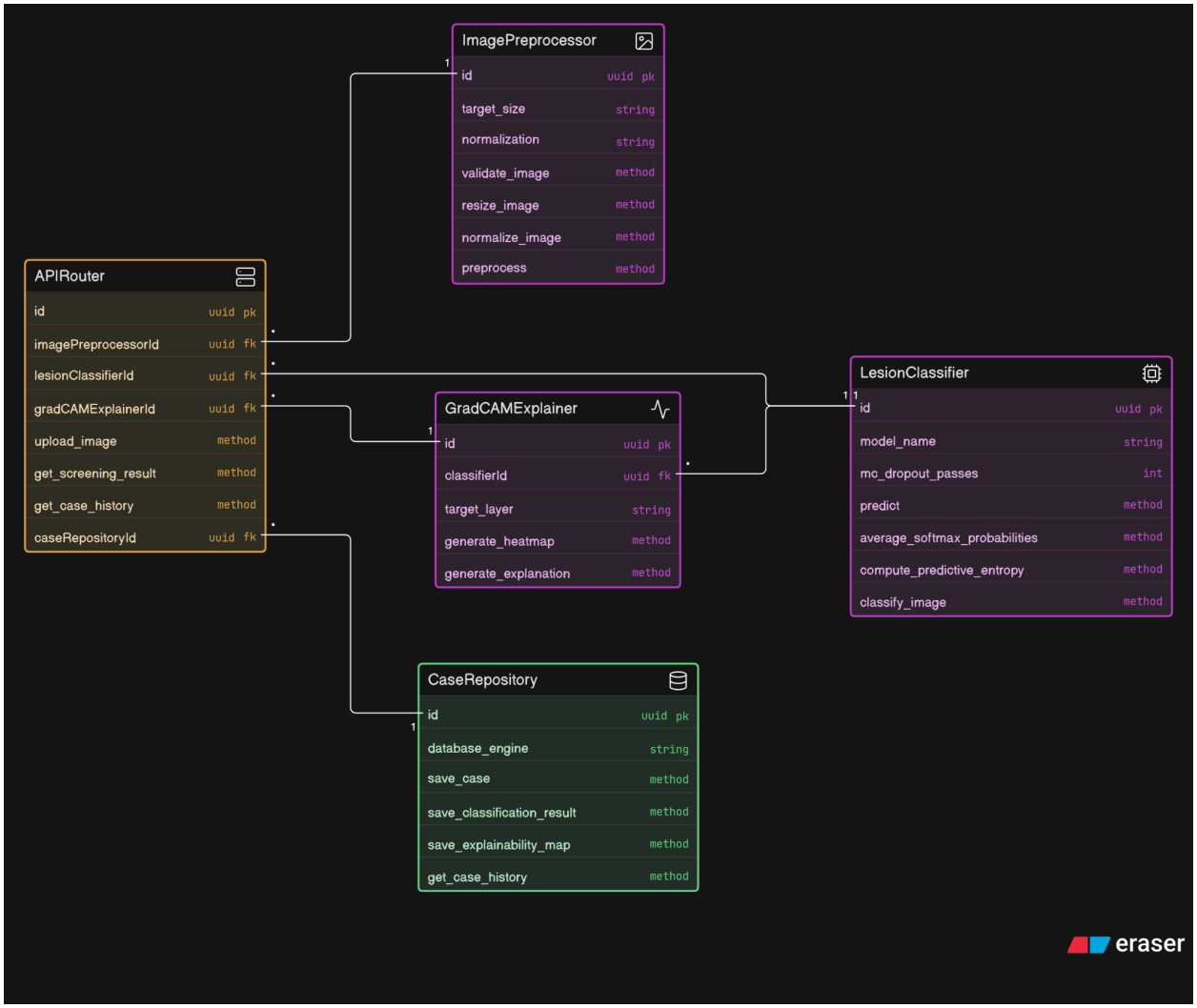


Figure 3.5 – Class Diagram: Principal classes ImagePreprocessor, LesionClassifier, GradCAMExplainer, CaseRepository, APIRouter with attributes and methods shown in standard three-compartment UML notation with association arrows.

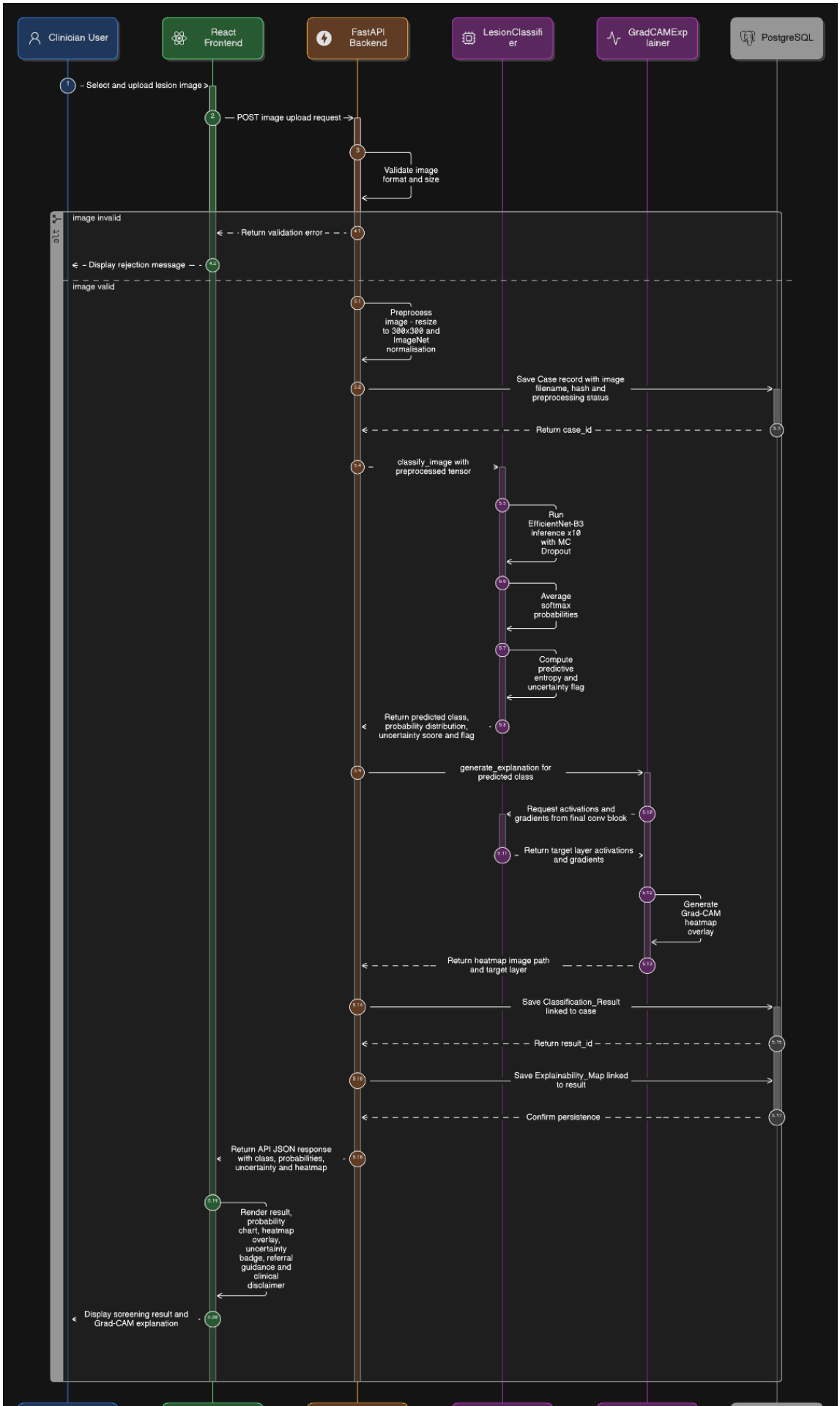

Clinician User
React Frontend
FastAPI Backend
LesionClassifier
GradCAMExplainer
PostgreSQL
1 – Select and upload lesion image
2 — POST image upload request
3 Validate image format and size
alt
image invalid
4.1 Return validation error
4.2 Display rejection message
image valid
5.1 Preprocess image - resize to 300x300 and ImageNet normalisation
5.2 Save Case record with image filename, hash and preprocessing status
5.3 Return case_id
5.4 classify_image with preprocessed tensor
5.5 Run EfficientNet-B3 inference x10 with MC Dropout
5.6 Average softmax probabilities
5.7 Compute predictive entropy and uncertainty flag
5.8 Return predicted class, probability distribution, uncertainty score and flag
5.9 generate_explanation for predicted class
5.10 Request activations and gradients from final conv block
5.11 Return target layer activations and gradients
5.12 Generate Grad-CAM heatmap overlay
5.13 Return heatmap image path and target layer
5.14 Save Classification_Result linked to case
5.15 Return result_id
5.16 Save Explainability_Map linked to result
5.17 Confirm persistence
5.18 Return API JSON response with class, probabilities, uncertainty and heatmap
5.19 Render result, probability chart, heatmap overlay, uncertainty badge, referral guidance and clinical disclaimer
5.20 Display screening result and Grad-CAM explanation

Figure 3.6 – Sequence Diagram (Image Upload Flow): Lifelines Clinician User, React Frontend, FastAPI Backend, LesionClassifier, GradCAMExplainer, PostgreSQL. Standard UML sequence notation (lifelines, activation boxes, synchronous messages, return arrows).

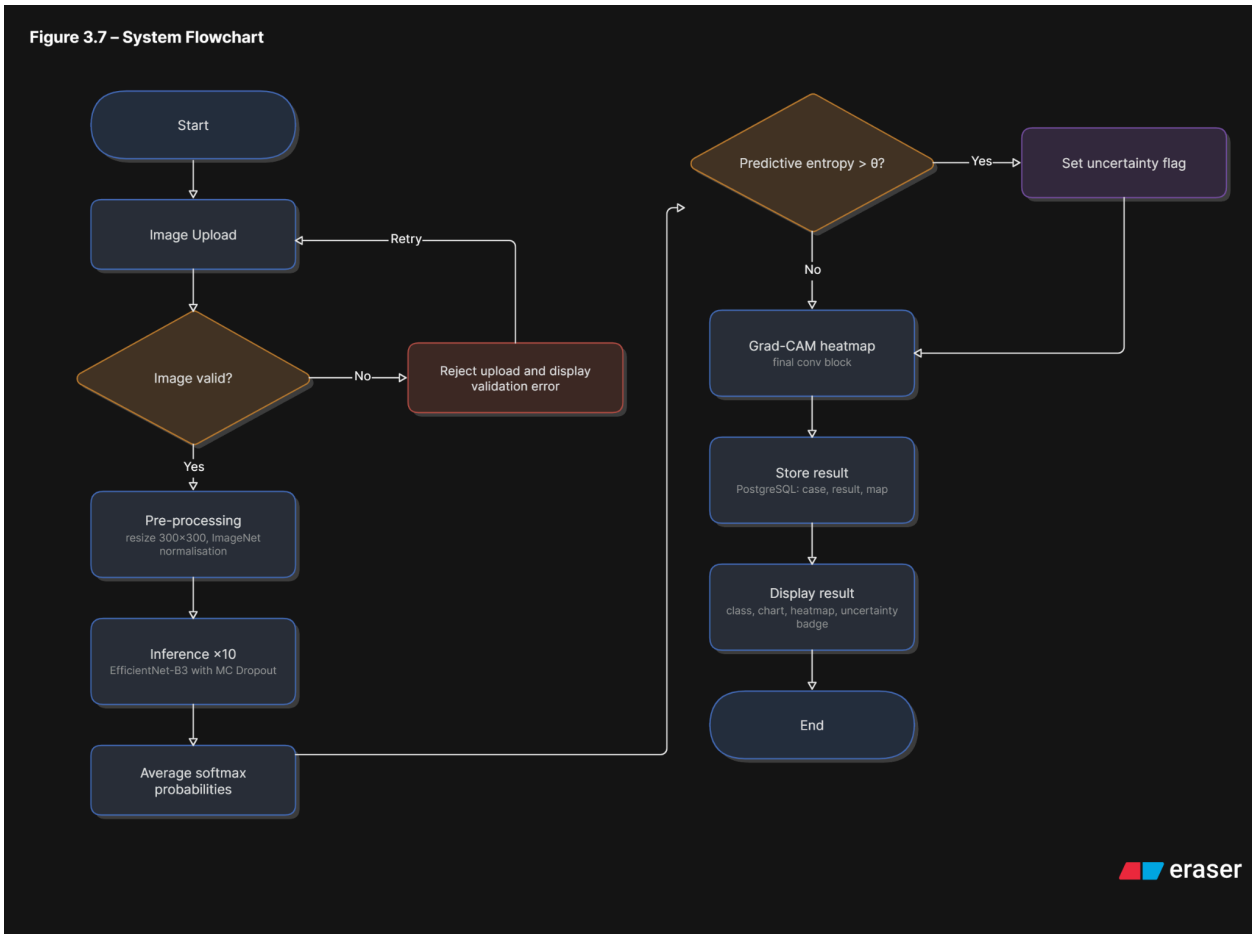


Figure 3.7 – System Flowchart

3.5 Development Tools

- Python 3.10+: Core backend and ML language.
- PyTorch 2.x together with torchvision: a deep learning framework, including pre-trained versions of EfficientNet-B3 and MobileNetV2 through torchvision.models.
- PyTorch-Grad-CAM: a version of Grad-CAM and Grad-CAM++ for PyTorch, which lessens the burden of implementing explainability.

- FastAPI is a Python framework for creating high-performance REST APIs and it includes automatic generation of OpenAPI documentation.
- React.js 18 and later is a frontend framework for clinician interfaces.
- PostgreSQL 15 combined with the SQLAlchemy ORM offers a relational database that has support for JSONB regarding probability distributions.
- Docker and Docker Compose: a means of achieving reproducible deployment through containerisation.
- Pillow and OpenCV: image pre-processing.
- pytest handles backend unit and integration tests.
- Git and GitHub (www.github.com/dalynahgasarasi) is a version control system and a public code repository.

References

Daneshjou R., Vodrahalli K., Novoa RA, Jenkins M, Liang W, Rotemberg V, Ko J, Swetter SM, Bailey EE, Gevaert O, Espinosa RL, Chiou AS and Zou J (2022) Disparities in dermatology AI performance on a diverse, curated clinical image set. Science Advances, 8(31), eabq6147. https://doi.org/10.1126/sciadv.abq6147

Doximity Foundation. (2023). Experiences initiating dermatological care in rural Rwanda. https://foundation.doximity.com/projects/1648

Esteva, A., Kuprel, B., Novoa, R. A., Ko, J., Swetter, S. M., Blau, H. M., & Thrun, S. (2017). Dermatologist-level classification of skin cancer with deep neural networks. Nature, 542(7639), 115–118. https://doi.org/10.1038/nature21056

Gal, Y., & Ghahramani, Z. (2016). Dropout as a Bayesian approximation: Representing model unpredictability in deep learning. Proceedings of the 33rd ICML, 48, 1050–1059.

Groh, M., Harris, C., Soenksen, L., Lau, F., Han, R., Kim, A., Koochek, A., & Badri, O. (2024). Deep learning-aided decision support for diagnosis of skin disease across skin tones. Nature Medicine, 30, 573–583. https://doi.org/10.1038/s41591-024-02848-4

Kinyanjui, N. M., Odonga, T., Cintas, C., Codella, N. C. F., Panda, R., Sattigeri, P., & Varshney, K. R. (2020). Fairness of classifiers across skin tones in dermatology. Proceedings of MICCAI, 320–329. https://doi.org/10.1007/978-3-030-59725-2_31

Leibig, C., Allken, V., Ayhan, M. S., Berens, P., & Wahl, S. (2017). Leveraging uncertainty information from deep neural networks for disease detection. Scientific Reports, 7, 17816. https://doi.org/10.1038/s41598-017-17876-z

Mahbod, A., Schaefer, G., Dorffner, G., & Ellinger, I. (2022). SkiNet: A deep learning framework for skin lesion diagnosis with uncertainty-estimation and explainability. Journal of Biomedical Informatics, 132, 104131.

Ministry of Health Rwanda. (2018). The National Digital Health Strategic Plan 2018–2023. Government of Rwanda.

Ministry of Health Rwanda. (2023). The 4x4 Reform: A path to quality health care in Rwanda. https://www.moh.gov.rw/news-detail/the-4x4-reform-a-path-to-quality-health-care-in-rwanda

Olubunmi, T., Adesanya, O., Bello, A., & Groh, M. (2025). Uncertainty-aware and explainable ensemble deep learning framework for multi-class skin lesion classification. arXiv:2608.11280.

Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L.-C. (2018). MobileNetV2: Inverted residuals and linear bottlenecks. Proceedings of IEEE CVPR, 4510–4520.

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient based localization. Proceedings of IEEE ICCV, 618–626. https://doi.org/10.1109/ICCV.2017.74

Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. Proceedings of the 36th ICML, 97, 6105–6114.

Tschandl, P., Rosendahl, C., & Kittler, H. (2018). The HAM10000 dataset. Scientific Data, 5, 180161. https://doi.org/10.1038/sdata.2018.161

World Bank. (2019). Physicians (per 1,000 people) – Rwanda. https://data.worldbank.org/indicator/SH.MED.PHYS.ZS?locations=RW